



SEMESTER 1 EXAMINATIONS 2024/2025

MODULE: CSC1028 - Probability & Statistics

PROGRAMME(S):

CASE	BSc Computer Apps (Sft.Eng.)
COMSCI	BSc Computer Science
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Program EC

YEAR OF STUDY: 1,2

EXAMINER(S):

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TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer 4 questions. All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1**[TOTAL MARKS: 25]**

A well-shuffled deck of cards contains 52 cards, comprised of 4 suits: 13 Diamonds, 13 Spades, 13 Clubs, and 13 Hearts. Within each suit, each card is labelled Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, and King.

For each of these questions **explain your answers** i.e. show your reasoning.

Q 1(a)**[14 Marks]**

Suppose we were to sample **with replacement** and draw N cards. What is the probability that:

- I. the first card is an Ace or a Spade?
- II. the second card is an Ace given that the first card was an Ace?
- III. the first card is an Ace, the second card is a Diamond, and the third card is a Diamond?
- IV. the first, third and fifth card are the same suit? e.g. all Diamonds or Spades or Hearts or Clubs.
- V. if 5 cards are drawn that each card will be unique?

Q 1(b)**[11 Marks]**

Suppose we were to sample **without replacement** and draw N cards.

- I. If I draw 5 cards from the deck (without replacement), what is the probability that 4 of these will be Diamonds?
- II. If I draw 5 cards from the deck (without replacement), what is the probability that at least one of the cards will be a Jack, Queen, or King?
- III. If I draw cards at random (without replacement), what is the probability the first Ace I draw will be the 5th card?

[End of Question 1]

QUESTION 2**[TOTAL MARKS: 25]**

For each of these questions **explain your answers** i.e. show your reasoning.

Q 2(a)**[13 Marks]**

Five delivery companies (numbered 1-5) are used by an organisation to send packages that contain documents. The table below indicates the fee charged by each delivery company to deliver a package, along with the total number of deliveries they have managed for the organisation.

Delivery company (ID)	1	2	3	4	5
Delivery fee per package (€)	2	4	8	11	2
Total number of deliveries	17	11	4	6	12

- I. What are the mean and variance of the expected delivery fee for the organisation?
- II. What is the probability that delivery company 1, 2 or 3 has been used, if we are told the delivery fee is greater than €5?

Q 2(b)**[7 Marks]**

The organisation uses the following formula to calculate the cost of preparing a package for delivery based upon the number of documents it contains:

$$\text{Cost} = 3X + 5$$

Where X = the number of documents in the package

We are told the average number of documents in a package is 9, with a variance of 3.

- I. What is the expected mean cost to prepare a package for delivery? i.e. find $E(3X + 5)$
- II. What is the expected variance of the cost to prepare a package for delivery? i.e. find $V(3X + 5)$

Q 2(c)**[5 Marks]**

We are told the probabilities for each delivery company losing a package are: 0.3, 0.4, 0.5, 0.6 and 0.7. If all 5 delivery companies are used to send the same copy of a document, what is the probability that between 1 and 4 copies of the document are delivered?

[End of Question 2]

QUESTION 3**[TOTAL MARKS: 25]**

A programmer writes programs in one of either Java, Fortran or Kotlin, where some of these programs contain errors. We are told 70% of all programs are written in Java, 10% in Fortran and 20% in Kotlin.

Examining only the programs that produce errors, we find 30% are written in Java, 20% in Fortran, and 50% in Kotlin.

We are also told that 40% of the programs written in Kotlin contain errors.

For each of these questions **explain your answers** i.e., show your reasoning.

Q 3(a) **[3 Marks]**

If we are told a program is written in Kotlin, what is the probability that it contains no errors?

Q 3(b) **[4 Marks]**

What is the probability a program is written in Java, if we are told that the program is not written in Kotlin?

Q 3(c) **[7 Marks]**

What is the probability if a program is chosen at random that it contains errors?

Q 3(d) **[6 Marks]**

What is the probability if a program is written in Java, that it contains no errors?

Q 3(e) **[5 Marks]**

What is the probability a program is written in Java and contains errors?

[End of Question 3]

QUESTION 4**[TOTAL MARKS: 25]**

In a garage open 24 hours a day, an average of 4 cars arrive each day. Of these cars, 25% are for servicing, while the remaining 75% need repairs. Cars are equally likely to arrive at any time of day or on any day of the week.

For each of these questions **explain your answers** i.e., show your reasoning.

Q 4(a) [2 Marks]

What is the average number of hours between the arrival of cars?

Q 4(b) [4 Marks]

What is the probability that no cars will arrive in a 36-hour period?

Q 4(c) [4 Marks]

What is the probability that 6 cars arrive for repairs on the same day?

Q 4(d) [4 Marks]

What is the probability of having to wait less than 4 hours for a car to arrive for a service?

Q 4(e) [5 Marks]

For what duration of time (in hours) would the probability of no cars arriving be 0.2636?

Q 4(f) [6 Marks]

What is the probability that the same number of cars will arrive at the garage for repairs in the next 6 hours as arrived in the previous 6 hours? Provide your answer to 2 decimal places.

[End of Question 4]

QUESTION 5

[TOTAL MARKS: 25]

Answer any 2 of the following questions i.e. of Q5(a), Q5(b) or Q5(c). There are equal marks for both questions.

For each of these questions **explain your answers** i.e. show your reasoning.

Q 5(a)

[12.5 Marks]

A production line produces devices consisting of four (4) types of components in a series system: C1, C2, C3 and C4. We are told there is a 0.1 probability that C1 is defective, a 0.2 probability that C2 is defective, a 0.3 probability that C3 is defective, and a 0.4 probability that C4 is defective. **note:** *in this series configuration, if a single component is defective, the whole device is defective.*

- I. What is the probability that a device is not defective?
- II. What is the probability that the first defective device from the production line will be the 5th device inspected?
- III. If each component is replicated in parallel, what is the probability that the system is defective?

Q 5(b)

[12.5 Marks]

The average shoe size is 8 with a standard deviation of 1.5. You can assume shoe sizes follow a normal distribution.

- I. What is the probability that the shoe size of a person is greater than 9.5?
- II. What is the probability that the shoe size of a person is greater than 8, if we are told their shoe size is less than 9.5?
- III. We are told that there is a 0.3085 probability that a person's shoe size is above a certain value. What is their shoe size?

Q 5(c)

[12.5 Marks]

In a computer installation, there are 5 servers. The probability that any given server is busy is 0.65.

- I. What is the probability that at a given moment none of the servers are busy?
- II. What is the probability that at a given moment more than 3 servers are busy?
- III. What is the expected mean and variance of the number of busy serves?

[End of Question 5]

APPENDICES

Area under the standard normal curve from $-\infty$ to z



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6338	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

${}^nC_k = \frac{{}^nP_k}{k!} = \frac{n!}{k!(n-k)!}$	
${}^nP_k = n(n-1)\dots(n-k+1) = \frac{n!}{(n-k)!}$	
$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$	
$P(A B) = \frac{P(A)P(B A)}{P(B)}$	$P(A \cap B) = \frac{P(A)P(B A)}{P(B)P(A B)}$
$V(X) = \sum_x (x - \mu_x)^2 p(x)$	$E(X) = \sum_x xp(x)$
$P(X = x) = \binom{n}{x} p^x q^{n-x}$	$P(X \leq x) = 1 - q^x$ $P(X = x) = q^{x-1} p$
$P(X = x) = \frac{\binom{M}{x} \binom{L}{n-x}}{\binom{N}{n}}$	$P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}$

[END OF APPENDICES]

[END OF EXAM]